

Assignment 1

Part I

Q1.

A traditional file system stores data in separate files and requires application programs to manage the data directly. A Database Management System (DBMS) is specialized software that stores, organizes, and manages data while providing features such as security, concurrency control, data integrity, and efficient querying.

A university student management system is a good example where a DBMS is clearly the better choice. Thousands of students, faculty members, and administrators may access the system simultaneously. A DBMS ensures data consistency, prevents duplication, and allows multiple users to work with the data safely and efficiently.

Q2.

A strong entity has its own primary key and can be uniquely identified without depending on another entity. A weak entity does not have a complete primary key of its own and depends on a related strong entity for identification.

The primary key of a weak entity is formed by combining the primary key of the strong entity with the weak entity's partial key. The relationship that connects the weak entity to its owner entity is called an identifying relationship.

Q3.

Phone number should be modeled as a multivalued attribute. A student may have more than one phone number, such as a personal number, home number, or guardian's number. Since multiple values can exist for a single student, a multivalued attribute is the most suitable representation.

Q4.

A super key is any set of attributes that uniquely identifies a tuple in a relation. A candidate key is a minimal super key, meaning no attribute can be removed from it while still maintaining uniqueness.

For the relation:

Student(roll_no, email, name, branch)

assuming both roll_no and email are unique:

Candidate Keys

- roll_no
- email

Super Keys that are not Candidate Keys

- (roll_no, name)
- (email, branch)
- (roll_no, email)
- (roll_no, name, branch)

Q5.

- **Constraint Violated:** Referential Integrity Constraint (Foreign Key Constraint)
- **What it Protects Against:** It prevents records from referencing non-existent rows in another table and ensures consistency between related tables.
- **What Should Happen:** The insertion operation should be rejected by the DBMS and an error should be generated because the referenced student does not exist.

Q6.

- **Additional Structure Required:** A separate junction table (also called an associative table) must be created.
- **Primary Key:** The primary key consists of the primary keys of the two participating entities combined.
- **Reason:** A many-to-many relationship allows multiple associations on both sides. Storing these values in a single column would violate normalization principles and create redundancy and update anomalies. Therefore, a separate table is required.

Part II

Part A: Entity and Relationship Identification

Entities and Key Attributes

1. Customer

- CustomerID (Primary Key)
- CustomerName
- Phone
- Address

2. Restaurant

- RestaurantID (Primary Key)
- RestaurantName

- Location

3. MenuItem

- MenuItemID (Primary Key)
- ItemName
- Price

4. Order

- OrderID (Primary Key)
- OrderDate
- TotalAmount

Relationships

1. Customer – Order (1:N)

One customer can place multiple orders, while each order is placed by exactly one customer.

2. Restaurant – Order (1:N)

A restaurant can receive multiple orders, while each order is associated with exactly one restaurant.

3. Restaurant – MenuItem (1:N)

A restaurant can offer multiple menu items, while each menu item belongs to one restaurant.

4. Order – MenuItem (M:N)

An order may contain multiple menu items, and a menu item can appear in multiple orders.

Part B: Attribute Classification

Multivalued Attribute: Customer Phone Number

A customer may have multiple phone numbers, such as a personal number and an alternate contact number. Since more than one value can be associated with a single customer, it is a multivalued attribute.

Composite Attribute: Customer Address

Address can be divided into meaningful subparts such as house number, street, city, state, and postal code. Therefore, it is a composite attribute.

Part C: Relational Schema Design

Since the relationship between Order and MenuItem is many-to-many, a junction table is required.

Relational Schema

OrderMenuItem(OrderByID, MenuItemID, Quantity)

Primary Key

(OrderByID, MenuItemID)

Foreign Keys

- OrderID $\rightarrow$ Order(OrderID)
- MenuItemID $\rightarrow$ MenuItem(MenuItemID)

Part III

Question 1

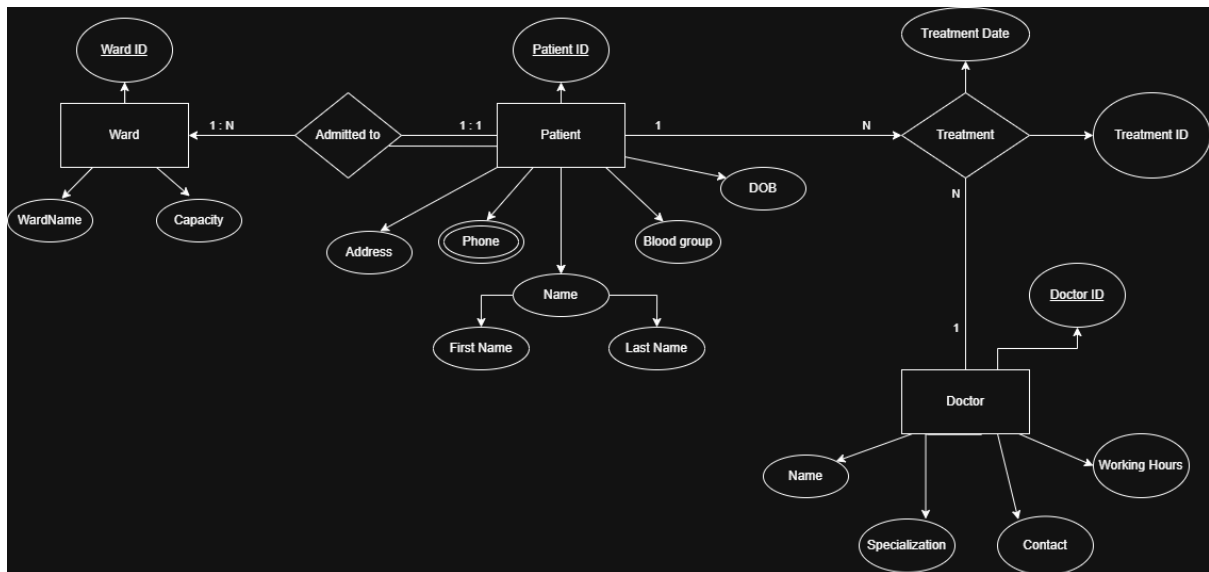


Figure 1: Hospital ER Diagram

Question 2

The Doctor–Patient relationship is many-to-many because a single doctor can treat multiple patients, while a patient may receive treatment from several doctors during a hospital stay. Since relational databases cannot directly represent many-to-many relationships, an associative entity called **Treatment** is created. The composite primary key of this entity consists of **PatientID** and **DoctorID**, ensuring that each doctor–patient association is uniquely identified.

Considering participation constraints, the relationship between Patient and Ward demonstrates total participation on the Patient side because every admitted patient must be assigned to a ward. On the other hand, Ward has partial participation because a ward may temporarily contain no patients. In the Doctor–Patient relationship, both entities have partial participation since a doctor may not currently be treating any patient, and a patient may be registered in the system before treatment begins.

One attribute design decision was modeling Address as a composite attribute. Instead of storing it as a single field, it was divided into components such as street, city, state, and postal code. This approach improves data organization, simplifies searching and filtering, and allows individual components of the address to be updated independently.